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Robotic optical coherence tomography with expanded three-dimensional field-of-view using point-cloud-based volumetric montaging

Raymond Fang,

Department of Biomedical Engineering, Northwestern University, Evanston, IL 60208

Pengpeng Zhang,

Department of Mechanical Engineering, Northwestern University, Evanston, IL 60208.

Tingwei Zhang,

Department of Biomedical Engineering, Northwestern University, Evanston, IL 60208

Zihang Yang,

Department of Biomedical Engineering, Northwestern University, Evanston, IL 60208

Daniel Kim,

Department of Biomedical Engineering, Northwestern University, Evanston, IL 60208

Edison Sun,

Department of Biomedical Engineering, Northwestern University, Evanston, IL 60208

Roman Kuranov,

Department of Biomedical Engineering, Northwestern University, Evanston, IL 60208

Opticent Health, Evanston, IL 60201, USA

Junghun Kweon,

Department of Biomedical Engineering, Northwestern University, Evanston, IL 60208

Alex S. Huang,

Department of Ophthalmology, University of California San Diego, San Diego, CA 92037.

Hao F. Zhang

Department of Biomedical Engineering, Northwestern University, Evanston, IL 60208

Abstract

Imaging complex, non-planar anatomies with optical coherence tomography (OCT) is limited by the optical field of view (FOV) in a single volumetric acquisition. Combining linear mechanical translation with OCT extends the FOV but suffers from inflexibility in imaging non-planar anatomies. We report the robotic OCT to fill this gap. To address challenges in volumetric reconstruction associated with the robotic movement accuracy being two orders of magnitudes worse than OCT imaging resolution, we developed a volumetric montaging algorithm. To test the robotic OCT, we imaged the entire circumferential aqueous humor outflow pathway, whose

imaging has the potential to customize glaucoma surgeries but is typically constrained by the FOV in mice *in vivo*. We acquired volumetric OCT data at different robotic poses and reconstructed the entire anterior segment of the eye. From the segmented Schlemm's canal volume, we showed its circumferentially heterogeneous morphology; we also revealed a segmental nature in the circumferential distribution of collector channels with spatial features as small as a few micrometers.

Index Terms—

Optical coherence tomography; robotics; simultaneous localization and mapping; aqueous humor; glaucoma

I. INTRODUCTION

Optical coherence tomography (OCT) has revolutionized fundamental investigations in various medical disciplines, including cardiology, dermatology, and ophthalmology[1–3]. In particular, OCT has transformed the clinical management for nearly all blinding diseases [4–7]. OCT is a non-invasive microscopic imaging modality that acquires volumetric data by detecting back-scattered photons at each optical illumination position and translating the optical focus of illumination to cover the region of interest (ROI). Such a data acquisition scheme has remained essentially unchanged since OCT was first reported in 1991.

All OCT systems have a finite imaging volume, defining the spatial coordinates that are observable by the design, which is related to the field of view (FOV) and the depth of the field (DOF). As illustrated in Fig. 1a, the optical implementation determines the FOV, and the DOF is governed by the optical penetration depth and spectral domain sampling density[8]. The limited OCT imaging volumes are often insufficient to cover large ROIs with spatial heterogeneity. Extending the imaging volume is achieved by implementing 3-degree-of-freedom (DoF) translational motions in Cartesian coordinates (Fig. 1b), as seen in nearly all modern microscopes[9]. Although this strategy enlarges the compound imaging volume (CIV) for stationary, flat samples, it faces several challenges in imaging non-planar anatomical structures. Existing hybrid scanning only allows a fixed, predefined optical illumination direction. Lacking the capability to adjust the optical illumination direction, especially for samples with non-planar features, true for nearly most structures *in vivo*, results in nonconformal imaging, where the incident illumination is oblique to the normal vector of the tissue surface. This nonconformal imaging reduces imaging accuracy due to degraded signal-to-noise ratio (SNR) and en-face projection artifacts. Hence, an ideal OCT should maximize the CIV, follow the sample contour for conformal scanning, and adjust the illumination direction as needed.

We developed a robotic OCT system to meet these needs. This system integrates an OCT sample arm with a six-axis robotic arm, allowing fully flexible, conformal scanning across non-planar sample geometries (Fig. 1c). The inclusion of rotational motion enhances the system's adaptability to maintain a normal OCT illumination to an arbitrary sample surface, thereby optimizing imaging quality.

Using a six-axis robotic arm offers flexibility in acquiring conformal CIV but introduces two primary technical challenges. First, the conformal CIV is specific to sample geometry, requiring knowledge of the sample geometries' local coordinates relative to the robotic arm's reference frames. Second, despite the high precision of the six-axis robotic arm, its accuracy is relatively limited [10–14], especially compared with the resolution of OCT. This discrepancy poses significant challenges in reconstructing CIVs by montaging multiple OCT volumetric images with an axial resolution of 2–5 μm . While the top commercial robots can achieve positioning precision of 5 μm , their absolute positioning accuracy typically exceeds 100 μm [15–18]. Bridging this gap of nearly two orders of magnitude between spatial accuracy and the spatial resolution of OCT presents an obstacle in reconstructing CIVs. Even though the positioning accuracy of 6-axis robots can be improved with high-precision metrology tools like coordinate measurement machines and laser interferometer trackers [19, 20], these approaches add unwanted burdens of technical complexity and cost, particularly in imaging biological samples.

Previous studies have demonstrated the capacity of robotic OCT to expand OCT's FOV [5, 18, 21–26]. However, owing to the aforementioned challenges, 3D rendering of the CIV remains unsolved. Huang et al. manually mosaiced individual sub-volumes together to create CIVs for *ex vivo* phantoms but noted boundary artifacts between sub-volumes and a limited robotic arm accuracy of around 0.21 mm [18]. Lotz et al. mosaiced the en-face projection of multiple sub-volumes to generate an enlarged lateral FOV of a human hand [21]. However, a 3D CIV integrating data from all sub-volumes was not achieved, where all cross-sections were aligned by intensity-based 2D registration of adjacent B-scans. Hence, substantial challenges remain in combining robotics and OCT to generate complex 3D anatomical structures.

To address these challenges, we repurposed the OCT as a built-in optical metrology tool, achieving two critical objectives simultaneously. Firstly, we delineated the distinct geometries of the samples, aligning the robotic arm's reference frames with the local coordinates of the samples. Secondly, we identified unique anatomical features on the samples using OCT, guiding the precise reconstruction of CIVs through the montage of multiple OCT volumetric images. We tested our OCT by imaging the anterior segment and aqueous humor outflow (AHO) pathway in mouse eyes.

We selected the AHO pathway for its anatomical complexities, making imaging the entire AHO pathway with traditional OCT difficult, and its clinical relevance to glaucoma. Glaucoma is a leading cause of irreversible blindness worldwide, and the leading and only modifiable risk factor for glaucoma is elevated intraocular pressure (IOP)[27]. The facility of aqueous humor outflow through the AHO pathways is critical for the final IOP magnitude and regulation. In both rodents and humans, the trabecular pathways are responsible for the majority of AHO. It is comprised of aqueous humor flowing from the anterior chamber to the trabecular meshwork (TM), Schlemm's canal (SC), collector channels (CC), and distal outflow vessels. [28] Pathological increases in resistance within the trabecular AHO pathways lead to elevated IOP. Visualizing the AHO pathway's anatomy in three dimensions (3D) at a microscopic spatial resolution can revolutionize glaucoma drug development [29] and enhance the efficacy of minimally invasive glaucoma surgeries (MIGS) in patients [30,

31]. However, imaging the AHO pathway presents substantial challenges using existing OCTs due to the AHO pathway's anatomical complexity (including structures as small as 2 μm), the orientation of those features oblique to the optical axis of the eye, and its circumferential distribution around the limbus. Additionally, CCs and distal pathways radiate outward, extending several millimeters beyond the limbus. Although anterior segment OCT has been used to visualize AHO pathways in humans [32], this imaging occurred in sampled locations. Our group has performed circumferential OCT imaging, but non-robotic commercial devices require over 5000 B-scans, days to image, and risk of dry eye [33]. Challenges in circumferential imaging of the AHO pathway and these distinctive anatomical characteristics underscore the necessity for a robotic OCT system.

We developed a robotic visible-light optical coherence tomography (vis-OCT) to image the entire AHO and its key components (TM, SC, and CC) within the CIV. Vis-OCT achieved an axial resolution of 1.3 μm , well suited for visualizing AHO pathways in mice. During the circumferential robotic scanning, we ensured that the vis-OCT illumination light remained conformal and perpendicular to the limbal surface. Our volumetric montaging achieved an accuracy around 10 μm , overcoming the intrinsic limit of the robotic arm's limited accuracy. This enabled the volumetric reconstruction of the entire AHO pathways, which further unveiled segmental variations in SC morphology and CC distribution and offered new insights into the structural complexity of the AHO system.

II. Materials and Methods

A. Robotic vis-OCT and data acquisition workflow

Fig. 2a shows the schematic of the robotic vis-OCT system. Broadband light is emitted from a supercontinuum laser (SuperK 78 MHz EXW-6, NKT Photonics). The output laser light passes through a dichroic mirror (DMSP650, Thorlabs), bandpass filter (FF01-560/94-25, Semrock), and a spectral shaping filter (Hoya B-460, Edmund Optics). The filtered output was coupled into a 90:10 fiber coupler (FC1, TW560R2A2, Thorlabs), with 10% output to the sample arm and 90% to the reference arm. The sample arm consists of a collimator (RC02APC-P01, Thorlabs), a 3-mm compact galvanometer (RC025547, ScannerMAX), and a 25-mm achromatic lens (Ac127-025-A, Thorlabs). The reference arm consists of 2 collimators (HPUC-2A3A-400/700-P-6AC-11, OZ Optics), BK7 dispersion glass (WG11010, Thorlabs), a polarization controller (FPC030, Thorlabs), and a variable attenuator (NDC-50C-4M, Thorlabs). In the sample arm, collimated light from the fiber coupler was scanned by a galvanometer and focused onto the sample. Back-scattered light from the sample passed through FC1 and interfered with the light from the reference arm in the 50:50 fiber coupler (FC2, TW560R5A2, Thorlabs). Two well-calibrated spectrometers (Blizzard SR, Opticent Inc.) detected the interferogram simultaneously for balanced detection[34]. The vis-OCT spectral range is 510 nm – 610 nm, and all data was acquired at an A-line rate of 75 kHz.

We designed a compact vis-OCT sample arm (Fig. 2b) and mounted it on a robotic arm (Meca500, Mecademic Inc.), which supports a payload of 0.5 kg with a positioning precision of 5 μm over a maximum reach of 225 mm[35]. The positioning precision of the Meca500 is among the highest for commercial robotic arms. We 3D-printed the sample arm

enclosure using carbon fiber-reinforced polycarbonate polymer materials to reduce the total weight to 0.29 kg. We integrated a pupil camera (outer diameter: 2 mm) and a time-of-flight (TOF) proximity sensor to the sample arm. The pupil camera provided the initial video guidance to locate the eye, and the TOF sensor prevented the sample arm from colliding with the eye.

Operating the robotic vis-OCT system requires managing two independent cartesian coordinate systems of the eyeball and the vis-OCT imaging volume (Fig. 2c). The eyeball coordinates are represented by the eye reference frame (ERF, X-Y-Z axes), where the Z-axis aligns with the eyeball's visual axis, and the origin is positioned at the center of the spherical surface approximating the cornea. The vis-OCT imaging volume's coordinates are represented by the tool reference frame (TRF, x-y-z axes). Here, the x-y plane coincides with the vis-OCT image plane. The orientation of the x-axis and y-axis is calibrated to align with the scanning directions of the galvanometer scanner. The origin of TRF is at the center of the vis-OCT FOV, with the z-axis pointing along the optical axis of the sample arm. For conformal imaging of the AHO pathways, it is important to accurately calibrate the 6-DoF coordinate transformation between the ERF and the TRF.

To perform conformal AHO pathway imaging after calibrating the coordinate transformation from TRF to ERF, we first rotated the TRF 60 degrees about either the ERF X or Y-axes, which is near the angle of the trabecular AHO pathway relative to the ERF. Then we adjusted the OCT sample arm to rotate the TRF 360 degrees around the ERF's Z-axis (path highlighted by the red-colored trajectory in Fig. 2c). For each sample arm position, the OCT illumination light along the z-axis of the TRF remained normal to the tissue surface, which is approximately parallel to the minor axis of an ellipse fit to SC at the limbus of the eye (Fig. 2d). We show an example of the robotic vis-OCT rotating around the eye in Supplemental Video 1.

B. Image acquisition protocol

Defining the ERF: We brought the eye into the OCT FOV and verified the eye position with the pupil camera. Next, we used a customized LabVIEW program to perform real-time image analysis of B-scans. We applied customized algorithms on the B-scans to align the z-axis of the TRF with the Z-axis of the ERF and the transverse origins of the TRF and ERF.

First, we parallelized the z-axis of the TRF and the Z-axis of the ERF, equivalent to aligning the pose of the TRF such that the xy-plane of the TRF was parallel to the XY-plane of the ERF. Using the hypothesis that the primary symmetry axis within each of the cornea B-scan images closely resembled the Z-axis in ERF, we sought to parallelize the symmetry axis in each real-time B-scan with the optical axis of the TRF. To parallelize the axes, we evaluated the relative orientation of the z-axis of the TRF and the Z-axis of the ERF along the x-axis using the slope of a line fit to the image intensity center of mass in the z-direction as a function of transverse position x . We calculated the center of mass as a function of position x with the following equation, where $B(z, x)$ is the intensity of the B-scan at depth z and position x :

$$COM(x) = \frac{\sum_z B(z, x)z}{\sum_z B(z, x)}. \quad (1)$$

The pose of the robot arm was adjusted until the absolute value of the slope of a line fit through $COM(x)$ was minimal. We minimized the absolute value of the slope of a line fit by $COM(x)$ using a proportional-integral control strategy [36, 37], with the error assigned to the current slope of the line and the output of a rotation around the y-axis in degrees relative to the current TRF. As a default, we set the proportional gain K_p as 0.05 and the integral gain K_i as 0.0005, both of which can be adjusted by the operator. We stopped adjusting the robotic arm's pose when the error was less than a preset threshold of 0.01 or a value set by the operator. We repeated the same process for the y-axis.

Next, we spatially overlapped the z-axis of the TRF with the Z-axis in ERF, equivalent to aligning the x and y origin of the TRF with the X and Y origin of the ERF. To determine the X and Y origin of the ERF, we applied a threshold to the cornea of the central B-scan and took the apex of the cornea to be the X and Y origin of the ERF. Next, we translated the TRF to move the apex into the center of the OCT FOV. After parallelizing and spatially overlapping the z-axis of the TRF and the Z-axis of the ERF, we set the origin of the ERF at a fixed distance underneath the TRF origin in the z-direction, with the distance set as the average radius of the mouse eye.

360-degree imaging: After defining the ERF, we rotated the TRF 60 degrees about the ERF X-axis. Afterward, we acquired eight vis-OCT volumes, with 45-degree rotations of the TRF around the ERF Z-axis between each acquisition along the trajectory illustrated in Fig. 2b. The operator made minor adjustments to the robot's orientation before each acquisition as needed. Each volume had a lateral FOV of 2.04 mm, 512 A-lines per B-scan, and 512 B-scans per volume. The sample arm was translated along the TRF z-axis to maintain the focus of the tissue, with signal intensity in the central B-scan, a fixed distance set by the operator from the zero-delay line, typically maximized. Additional features besides the TOF sensor to prevent collisions included an emergency stop button and a constraint of the maximum velocity of the robotic motion of 10 mm/s. We applied a temporal speckle averaging data acquisition scheme [38], where each B-scan was acquired twice per sub-volume, and three sub-volumes were acquired per volume. Each volumetric acquisition took 25.1 seconds, with robotic motion and operator validation taking roughly 30 seconds between volumes.

C. Volumetric Montaging for OCT

We developed a volumetric montaging algorithm to reconstruct a CIV of the anterior segment of the mouse eye from the eight acquired volumes. For each volume, we recorded the orientation of the TRF relative to the ERF and reconstructed both the vis-OCT microanatomy and vis-OCT angiogram (Fig. 3a). After we determined which volumes overlapped with each other based on the recorded TRF orientations, we registered and montaged adjacent volumes to reconstruct the entire CIV. To montage adjacent volumes, we segmented the tissue surface and represented the outer surface as a point cloud [39]

(step 1 in Fig. 3b). We identified common vessel branch points between adjacent volumes in the angiogram, extracting the outer surface at the lateral position of the branch points (step 2 in Fig. 3b). Then, we used the M-estimator Sample Consensus algorithm [40] to determine the rigid geometric transformation between the common branch points (step 3 in Fig. 3b). All point clouds were transformed into the ERF coordinate system. Following the transformation into the ERF, we found the overlapping regions of the point clouds (Fig. 3c) and used the iterative closest point algorithm [41] to refine the transformations of each reference frame relative to the ERF.

Fig. 4a shows the steps of volumetric montaging.

Representation of the outer eye surface as point cloud: For each vis-OCT volume, we represented the outer surface of the eye closest to the incident OCT beam as a point cloud. We define $I(x, y, z)$ as the intensity of the OCT volume at coordinate (x, y, z) in the TRF (Fig. 4b). First, we binarized each B-scan by thresholding each B-scan, using the 15th percentile of image intensity as an automated threshold, and keeping the largest spatially connected object (Fig. 4c). We took the union of all the binarized B-scans to form a binarized volume (Fig. 4d). For every A-line in the binarized volume, we took points at the outer surface of the segmented volume and represented this set of points as a point cloud (Fig. 4e).

Montaging of overlapping point clouds: To determine whether two volumes A and B overlap, we checked whether the coordinates of the voxels for A and B had greater than 20% volumetric overlap in the ERF. An overlap of 20% is slightly less than the expected overlap area between acquired volumes, which ranges from 23 to 43%, assuming an eye diameter ranging from 3 mm to 4 mm and a lateral field of view of 2.04 mm. We note that the percentage of overlap between adjacent volumes depends on the acquisition scheme, but the overlapping volume must provide adequate shared landmarks between the volumes. The robotic pose of the TRF relative to the ERF during the acquisition of each volume was used to estimate the coordinates of the voxels for each volume. If the volume of the intersecting regions was greater than 20% that of volumes A and B, we considered volumes A and B to be overlapping. When A and B had sufficient overlap, we merged the point clouds of A and B.

We let P_A and P_B be the point cloud for volumes A and B. Furthermore, we let S_A and S_B be the functions returning the z coordinate of the point cloud at position (x, y) . In other words, for an arbitrary surface N, if the z coordinate of the outer surface of the segmented volume at (x_i, y_j) is z_{ij} , then

$$S_N(x_i, y_j) = z_{ij}. \quad (2)$$

We manually identified the lateral position of 15 different common landmark points, usually vessel branch points in the OCTA, in the overlapping region between A and B (Fig. 4f-g).

Supposing that the landmark points have lateral positions $\{(x_1, y_1), \dots, (x_{15}, y_{15})\}$ in volume A in the ERF, we define the set of landmark points L_A as

$$L_A = \{(x_1, y_1, S_A(x_1, y_1)), \dots, (x_{15}, y_{15}, S_A(x_{15}, y_{15}))\} \quad (3)$$

We define the landmark points L_B in volume B in the same fashion. We applied the M-estimator sample consensus algorithm to estimate the rigid transformation T_{AB} mapping L_A onto L_B (Fig. 4h) [40]. Next, we determined the points in the overlapping regions between $T_{AB}(P_A)$ and P_B (Fig. 4i). For each point in $T_{AB}(P_A)$, we found the closest point in P_B . We defined the set of closest points in P_B to $T_{AB}(P_A)$ as the region of P_B overlapping $T_{AB}(P_A)$, which we denote O_{BA} . We found the overlapping region O_{AB} , the set of closest points in P_A to $T_{AB}^{-1}(P_B)$, similarly. Then, we updated T_{AB} as the transformation mapping O_{AB} to O_{BA} using the iterative closest point (ICP) algorithm [39], with the initial estimate for the transformation the current transformation mapping L_A onto L_B . For our ICP algorithm, we used the point-to-point variant, considered all sets of matched points as inliers, and set the maximum number of iterations at 30. We set the stopping criteria as when the absolute difference between translations was less than 0.01 μm and the absolute difference between rotations was less than 0.5 degrees on average for three consecutive iterations.

After calculating the rigid transformations mapping each vis-OCT volume with its overlapping neighboring volumes, we determined the transformation mapping each vis-OCT volume to the ERF. First, we choose a volume to serve as an initial stationary volume. For discussion purposes, let $T_{A,ERF}$ be the relative pose of the TRF to the ERF when volume A was acquired. For volume A, the transformation mapping the coordinates of A to the ERF is $T_{A,ERF}$. If $T_{B,A}$ maps volume B to volume A, then $T_{B,A} * T_{A,ERF}$ would map the coordinates of volume B to the ERF.

After determining initial rigid transformations mapping each volume to the ERF, we applied the corresponding transformations to map the point cloud for each volume to the ERF. Next, we used the M-estimator Sample Consensus algorithm to fit a sphere to the union of all the volumetric point clouds in the ERF, to find the centroid of the fitted sphere, and to determine the inlier points used for the sphere fitting. Using the inlier points, we found a rotational axis of symmetry representing the visual axis of the eye. After determining the centroid of the fitted sphere and the visual axis of the eye, we updated each of the initial transformations. Specifically, if T_{Origin} translates the origin of the fitted sphere to the origin of the ERF and T_{Axis} aligns the calculated visual axis with the current Z-axis of the ERF, we updated the rigid transformation for volume A to the ERF as

$$T_{A,ERF} \leftarrow T_{Axis} T_{Origin} T_{A,ERF}. \quad (4)$$

We updated transformations for the other volumes identically.

Loop closure problem: When a series of transformations are applied consecutively, errors from each transformation cascade, resulting in mismatching boundaries for the CIV.

We mapped all overlapping regions into the ERF, showing the example mapping for O_{AB} below.

$$O_{AB} \leftarrow T_{A,ERF} O_{AB} \quad (5)$$

We defined O_{ERF} as the union of all overlapping regions between every overlapping volume. After mapping all overlapping regions to the ERF, we refined the transformation mapping each volume to the ERF. To optimize the transformations, we minimized the cost function, the sum of the root mean square error between the overlapping regions of each volume and O_{ERF} .

$$RMS(T_{1,ERF}, \dots, T_{8,ERF}) = \sum_{N=1}^{N=8} \frac{1}{|O_N|} \sum_{n \in O_N} \left\| T_{N,ERF}(n) - (O_{ERF} - T_{N,ERF}(O_N)) \right\|^2 \quad (6)$$

Here, O_N is the point cloud of overlapping regions for volume N , n is an individual point in O_N , $|O_N|$ is the number of points in O_N , and $\left\| T_{N,ERF}(n) - (O_{ERF} - T_{N,ERF}(O_N)) \right\|^2$ is the Euclidean distance between $T_{N,ERF}(n)$ and the closet point in $(O_{ERF} - T_{N,ERF}(O_N))$.

To generate an updated estimate of $T_{B,ERF}$, we determined the transformation mapping of the overlapping regions of B to the union of all overlapping regions. For this step, all point clouds were downsampled randomly to 25%. Supposing volume B overlapped with volumes A and C, we found the set of overlapping regions O_{BA} and O_{BC} . Then, we utilized the ICP point algorithm, using the same parameters as in the montaging adjacent point clouds, to calculate the transformation $T_{OB,OERF}$ mapping O_{BA} and O_{BC} onto $O_{ERF} - (O_{BA} \cup O_{BC})$ [39]. If $T_{B,ERF}$ initially mapped volume B to the ERF, then we updated the following:

$$T_{B,ERF} \leftarrow T_{OB,OERF} * T_{B,ERF}, \text{ and} \quad (7)$$

$$O_{BA} \leftarrow T_{OB,OERF}(O_{BA}), O_{BC} \leftarrow T_{OB,OERF}(O_{BC}).$$

(8)

We applied this process to every volume consecutively to refine the transformation of all volumes to the ERF (Fig. 4j). Following the sequential application of ICP to every volume, we recalculated RMS. We repeated this process until RMS decreased by less than 1% of its initial value before sequential ICP or a maximum of 10 iterations. As an optional additional step, we utilized the quasi-newton algorithm [42] to minimize the RMS, with the initial estimates of the input transformations coming from the sequential application of ICP. This optimization step reduces to a 48 degree-of-freedom minimization problem, with 6 degree-of-freedoms per volume. We set the termination tolerance of the quasi-newton algorithm as 0.001 and the maximum number of iterations as 200. Following transformation refinement, the union of all point clouds mapped onto the ERF was smooth and continuous (Fig. 4k).

Volumetric montaging: After identifying the rigid transformations T_A mapping P_A from its original image reference frame to the TRF for every volume, we applied the transformation to every volume. If T_A maps point (x, y, z) from the original image coordinate system of A to point (i, j, k) in the ERF, then:

$$I_{ERF}(i, j, k) = I_A(x, y, z). \quad (9)$$

When multiple points from different volumes are mapped onto the same point in the ERF, we took the maximum intensity of the volumes mapping onto that point. In other words, if T_A maps (x_a, y_a, z_a) to (i, j, k) and T_B maps (x_b, y_b, z_b) to (i, j, k) , then $I_{ERF}(i, j, k)$ would equate to the maximum of $I_A(x_a, y_a, z_a)$ and $I_B(x_b, y_b, z_b)$. In the reconstruction of I_{ERF} , we assigned a voxel size of $4\mu\text{m}$ in each dimension. We used linear interpolation to assign intensities to coordinates between the spatial locations of multiple voxels in the original volumes.

Representation of data within spherical coordinates: We transformed the reconstructed 3D volume from cartesian to spherical coordinates using the following transformation

$$I_{spherical}(r, \theta, \phi) = I_{ERF}(r \sin \theta \cos \phi, r \sin \theta \sin \phi, r \cos \theta). \quad (10)$$

En-face projections E were created as a function of θ and ϕ by

$$E(\theta, \phi) = \frac{\sum_{r=r_-}^{r_+} I_{spherical}(r, \theta, \phi)}{\sum_{r=r_-}^{r_+} 1}. \quad (11)$$

As a spherical surface cannot be perfectly projected onto a plane, we applied the various projections from cartography to $E(\theta, \phi)$ to render the eye in 2D (Fig. 41)[43].

D. Data processing and measuring Schlemm canal parameters

We reconstructed the anatomical vis-OCT images following methods in our previous work[44]. Further, we processed the angiography data using published OMAG and SSADA algorithms [45, 46]. To reduce the influence of bulk motion on OCT angiography, we performed image registration between repeated B-scans [47]. After reconstructing each volume, we mapped all volumes into the ERF. Additionally, we segmented the entire SC around the globe. For segmentation, we selected the lumen corresponding to SC in each B-scan of every volume and applied a flood fill algorithm (intensity within 20th percentile of image intensity to intensity of selected points) to identify the borders of the lumen. We manually checked and applied corrections to all segmented B-scans using MATLAB's (2022) Volume Segmenter application. By summing all SC cross-sectional areas within each quadrant of the eye, we calculated the regional volume per quadrant of the eye. Once SC was segmented, SC height and width corresponded to the minor and major axis dimensions of an ellipse fit to SC [48].

E. Animal care and handling

We anesthetized all mice by intraperitoneal injection (10 mL/kg body weight) of ketamine/xylazine cocktail (ketamine 11.45mg/L, xylazine 1.7mg/L in saline). While under anesthesia, the mouse's body temperature was maintained using an infrared lamp. To expose the entire 360 degrees of the limbus, we made a small incision to the upper and lower eyelids and inserted a speculum underneath the eyelids. We imaged the entire outflow pathway in six C57BL/6J mice and one BALB/c mouse, the partial outflow pathway in one additional C57BL/6J mouse, and the cornea in one C57BL/6J mouse. We imaged one eye per mouse and applied eye drops (NDC 57896–181-05, Gericare) before and after image acquisition. All experimental procedures were approved by the Institutional Animal Care and Use Committee at Northwestern University.

F. Phantom Imaging and Reconstruction

We 3D-printed a hemispherical phantom using our custom-built continuous liquid interface process (CLIP) printer with a polyethylene glycol diacrylate-based resin [49]. The phantom consists of a 4-mm diameter hemisphere adorned with cube-shaped elements ($150 \times 150 \times 150 \mu\text{m}$) that are evenly distributed every 20° on the surface.

We acquired eight volumes of the periphery of the phantom following the same protocol in mouse eye imaging, with the phantom being stationary during imaging. We reconstructed the phantom with the methodology presented above with one slight difference: instead of using OCTA to determine landmark points, we manually selected landmark points from the structural en face projection.

G. Statistical analysis

All statistical comparisons, unless otherwise specified, were done after performing a one-way ANOVA using GraphPadPrism 9.5.1. When multiple comparisons were made, we

performed a Tukey multiple comparison correction. All presented values are given by mean $\pm$ standard deviation unless otherwise specified.

III. Results

A. Influence of vis-OCT illumination incident angle

A key advantage of robotic OCT is its capability to adjust incident angle as needed. We examined the influence of illumination incident angle in imaging AHO pathways based on the contrast-to-noise ratio (CNR). Since the axial resolution of nearly all OCT is several folds better than its lateral resolution, anatomical features are best resolved axially instead of laterally [50]. Hence, maintaining a normal illumination can maximize the benefit of vis-OCT's high axial resolution in imaging fine anatomical features in the AHO pathway, such as the SC. If we simplify the cross-section of the SC as an ellipse, where the minor axis corresponds to its smaller dimension and the major axis to its larger dimension, a normal OCT illumination to the limbus will align with the minor axis to best resolve SC.

We imaged the SC (Fig. 5a) and the cornea (Fig. 5b) using two vis-OCT illumination incident angles: normal to the major axis of SC (highlighted by the red arrows) and to the central cornea (highlighted by the blue arrow). When the vis-OCT illumination aligns with the normal axis of the cornea, it was difficult to resolve the borders of SC (blue box in Fig. 5c) and hard to visualize the limbal vasculature from the vis-OCT angiography or vis-OCTA (Fig. 5d). We differentiated the corneal layers in the central cornea (blue box in Fig. 3e) but not the peripheral cornea (red box in Fig. 5e). When the vis-OCT illumination aligns with the minor axis of SC, we well resolved the borders of SC (red box in Fig. 5f) and the limbal vasculature network from vis-OCTA (Fig. 5g). Additionally, we observed different corneal layers in the peripheral cornea (red box in Fig. 5h) but not the central cornea (blue box in Fig. 5h).

Quantitatively, the CNRs between SC and surrounding tissue were 9.0 ± 1.8 when the vis-OCT beam was normal to the central cornea and 51.0 ± 24.7 when orientated along the minor axis of SC (Fig. 5i). Measurements of CNRs between SC and surrounding tissue were taken from B-scans located $119.5 \mu\text{m}$ apart. The CNRs between the epithelium and stroma in the central cornea were 19.9 ± 8.2 when orientated normal to the central cornea and 14.3 ± 6.1 when orientated along the minor axis of SC (Fig. 5j). The CNRs between the epithelium and stroma in the peripheral cornea were 9.5 ± 2.5 when orientated normal to the central cornea and 35.5 ± 14.4 when orientated along the minor axis of SC (Fig. 5k). Measurements of CNRs between epithelium and stroma were taken from B-scans located $159.4 \mu\text{m}$ apart. In conclusion, best imaging quality requires vis-OCT illumination beam to be normal to the sample surface, requiring conformal scanning.

B. Reconstructing the anterior segment

Next, we applied the calculated geometric transformations to each vis-OCT volume and visualized the reconstructed anterior segment from the anterior (Fig. 6a) and posterior (Fig. 6b) perspectives. A sample cross-section located $560 \mu\text{m}$ from the center of the eye shows a smooth corneal and iris surface, indicating accurate montaging (Fig. 6c). We transformed the

coordinate system of the reconstructed volume from a cartesian coordinate into a spherical coordinate system and identified the mean structural and angiographic signal as a function of θ and ϕ . Following coordinate transformation, we projected the structural and angiographic volumetric data into two dimensions using a stereographic projection (Figs. 6d-6e) [43]. Features, including the trajectory of individual blood vessels along the projections, are smooth and continuous between the montaged volumes. We show an example fly-through of the cross-sectional B-scans of the CIV in Supplemental Video 2.

To evaluate the volumetric montaging, we calculated the distance between point clouds from adjacent volumes. We calculated the distance between point clouds as the mean distance between each point in the overlapping regions of the point cloud of one volume to the closest point from another volume. For seven mice, each with eight volumes, we found that the mean distance between every point in the overlapping region for a volume and the closest point on the adjacent point cloud was $11.2 \pm 6.4 \mu\text{m}$ ($n = 56$ volumes). When the sequential ICP was used for loop closure and the optional minimization step using the quasi-Newton search algorithm was excluded, the mean distance increased to $11.9 \pm 6.6 \mu\text{m}$ ($n = 56$ volumes).

The total processing time for volumetric montaging was 15.0 ± 6.7 minutes and 4.6 ± 1.3 minutes when the optional minimization step was excluded. Computations were performed with a Core i7–10700k (Intel) processor without GPU usage. We documented the time consumption for each step in Table 1 ($n = 7$ mice).

C. Segmental pattern of Schlemm canal morphology

From the reconstructed anterior segment, we segmented the SC around the entire globe (Fig. 7a), which forms a continuous lumen (Fig. 7b). We approximated the cross-section of SC by an ellipsoid, with the major and minor axes corresponding to the SC width and height, respectively and selected several planes to visualize SC around the 360 degrees of the eye. When visualized along a plane passing through major axes of each cross-section of SC, SC forms a continuous hypo-reflective ring immediately lateral to the anterior chamber (Fig. 7c). Furthermore, we visualized segmental SC height patterns after digitally resampling [51] the anterior segment along a plane passing through the lateral center of SC (Fig. 7d). Overall, SC height is larger in the nasal and temporal quadrant for a C57BL/6J mouse. Although generally smooth, we note some montaging artifacts in Figs. 7c and d. These relate to volumetric registration inaccuracies and signal intensity differences between adjacent volumes. For each mouse, we measured segmental patterns in SC area, height, and width as a function of angle around the globe (Figs. 7e-g).

Relative to the mean volume across all quadrants for each eye, SC volume was $32.6 \pm 11.7\%$ larger in the temporal quadrant, $23.8 \pm 22.2\%$ smaller in the superior quadrant, $12.0 \pm 19.9\%$ smaller in the nasal, and $1.2 \pm 13.1\%$ larger in the inferior quadrant. Volume in the temporal quadrant was greater than the other quadrants (Fig. 7h). The average SC cross-sectional area for each quadrant was $2686.9 \pm 983.5 \mu\text{m}^2$, within the range of previous *in vivo* measurements [44, 52, 53]. Relative to the mean height across all quadrants, SC height was $21.0 \pm 7.4\%$ larger in the temporal quadrant, $16.7 \pm 15.2\%$ smaller in the superior quadrant, $6.6 \pm 13.8\%$ smaller in the nasal, and $0.9 \pm 7.7\%$ larger in the inferior quadrant. Height in

the temporal quadrant was greater than the other quadrants (Fig. 7i). Relative to the mean width across all quadrants, SC width was $10.7 \pm 7.5\%$ larger in the temporal quadrant, $9.5 \pm 9.3\%$ smaller in the superior quadrant, $2.8 \pm 8.3\%$ smaller in the nasal, and $0.9 \pm 6.1\%$ larger in the inferior quadrant. The width in the temporal quadrant was greater than the nasal and superior quadrants (Fig. 7j).

D. Segmental distribution of collector channels

As the segmental distribution of CCs has been hypothesized to influence MIGS outcomes [54], we visualized each CC around the globe in mice from the reconstructed anterior segment. We mapped the positions of CCs around the eye and correlated locations with the local SC area in C57BL/6 mice (Fig. 8a). We found that the mean total number of CCs was 33.7 ± 6.5 . The average number of CC in the inferior, nasal, superior, and temporal quadrants are 6.2 ± 2.0 , 10.5 ± 2.8 , 6.3 ± 2.3 , and 10.7 ± 2.6 respectively (Fig. 8b). Overall, the nasal and temporal quadrants contained more CCs than the inferior and superior quadrants ($p < 0.05$).

After digital resampling, we visualized every single CC in a C56BL/6 mouse (Fig. 7c). For each resampled CC image, we highlighted the connection between SC (blue arrows) and CC (green arrows). Several different CC morphologies were observed, with some CCs open throughout (#30 in Fig. 8c), some CCs exhibiting pinch points (#6 in Fig. 8c), some CCs exiting at a vertical angle (#22 in Fig. 8c), and some CCs exiting at a side angle (#33 in Fig. 8c). These results further validated the complexity of AHO pathway and the need for conformal OCT imaging.

E. Phantom validation

To further evaluate our robotic vis-OCT, we reconstructed a spherical phantom with cuboidal features evenly distributed at every 20° (Fig. 9a). Following the same acquisition protocol used in mouse eye imaging, we montaged the point clouds (Fig. 9b) and reconstructed the OCT CIV (Fig. 9c) of the phantom, with all cuboidal features oriented 40 to 80° relative to the Z-axis of the hemisphere included in the reconstruction.

We analyzed a horizontal plane passing through the surface of the hemisphere orientated 60° relative to the Z-axis of the hemisphere. In that plane, the X and Y coordinates of the CIV point cloud are shown in red and the X and Y coordinates of the designed phantom in green (Fig. 9d). For each of the vertices of the cuboidal features ($n = 36$ points, green arrow in Fig. 9d), we found the closest point in the point cloud (red arrow in Fig. 9d) to each of the vertices and calculated the distance between the point cloud and the location of the vertices in the theoretical design. We found that the mean distance was $8.0 \pm 7.0 \mu\text{m}$. When the Meca500's recorded poses, instead of our volumetric montaging scheme, were used to montage the hemisphere, the mean distance was $130.9 \pm 99.0 \mu\text{m}$. Since the matching points to each vertex were located slightly differently for each single volume, we measured the distance between points corresponding to each vertex from each volume. For adjacent volumes sharing vertices, we calculated the distance between the point closest to the vertex from the first volume from the point nearest to the point from the second volume. The mean distance between these points was $14.9 \pm 4.2 \mu\text{m}$.

We evaluated our method for determining the ERF Z-axis by determining the optical axis of the anterior chamber of a commercial phantom eye model (Rowe Technical) [55]. We manually determined the optical axis of the phantom eye model and intentionally misaligned the TRF z-axis with the optical axis between -5 to 5 degrees in increments of 1 degree. We applied our methodology to align the TRF z-axis with the phantom optical axis, with the difference between the TRF orientation and the estimated ground truth to be 0.21 ± 0.20 degrees, with a maximum difference of 0.68 degrees.

IV. Discussion

Limitations in mechanical scan angle and illumination orientation constrain the imaging volume for most OCT systems. Here, we present a robotic OCT capable of eye imaging to overcome these limitations. Although robotic OCT has previously been used to image the retina and expand the optical FOV, substantial challenges remain in generating 3D CIVs [5, 18, 21–26, 56]. To address these challenges, we developed a volume OCT registration algorithm, generating a CIV that encompasses the entirety of the AHO pathways in mice.

Compared with stationary OCTs, our robotic OCT has complete flexibility in controlling illumination incident angles during data acquisition. We found that CNR in imaging the AHO and cornea depended on OCT's illumination incident angle. Optimal imaging of complex non-planar tissues occurs when the OCT illumination beam is normal to the tissue surface. In our case, we accomplished this by calibrating the ERF relative to the TRF and rotating about the z-axis of the ERF. Using conformal scanning, we resolved the SC with much higher CNRs than could be achieved with a fixed beam angle. One advantage of conformal scanning is that the sample does not need to be repositioned to achieve normal illumination. Future work may include dynamic adjustment of angle based on the relative orientation of the robot and sample. Toward this objective, other investigators have used face and pupil cameras to automatically align the retina within the OCT FOV [57] and applied linear fitting of the sample surface to adjust orientation [18, 21]. For *in vivo* imaging, conformal scanning will allow fewer motion artifacts and improved reproducibility [58–60].

Following acquisitions of OCT volumes from multiple poses around the eye, we represented the surface of each volume as a point cloud and volumetrically montaged all the volumes. In traditional robotics, mapping algorithms integrating robotic positional data with images of the environment have been used to map out unknown environments [61]. In most cases, robotic positional sensors are more accurate than the surrounding objects being reconstructed. In our case, the resolution of OCT is several folds higher than the accuracy of robotic arms. Our vis-OCT has an axial resolution of $1.3 \mu\text{m}$ while the accuracy of most precise robotic arms exceeds $100 \mu\text{m}$ [35]. Consequently, robotic positional sensors cannot reliably merge all vis-OCT volumes. To address this issue, we developed a montaging method integrating robotic positional data with acquired OCT volumes to reconstruct the anterior segment of the eye. Robotic poses were only used to determine whether two volumes have sufficient overlapping regions. The fine montaging of 3-D structural features was accomplished by optimizing the geometric transformations mapping the OCT landmark points of adjacent point clouds onto each other with around $10 \mu\text{m}$ distance between landmark points in adjacent volumes, an order of magnitude better than the accuracy of

the robotic arm. The outer surface of the merged volumes and 360-degree reconstructed SC was smooth and continuous, indicating the accuracy of volumetric montage. For example, the montaged OCTA presented in Fig. 6E contains 8 regions consisting of overlapping volumes and 8 regions without overlapping volumes. The CNR between the vessels in the overlapping regions was 1.89 ± 0.59 , and the CNR in the non-overlapping regions was 1.94 ± 0.45 . Quantitatively, the contrast of structures in the regions consisting of multiple volumes is comparable to regions without multiple volumes.

The integration of mechanical design and novel processing expands the CIV of OCT, enabling the reconstruction of complex non-planar anatomies. Nearly all existing OCT montage algorithms register en-face projection images [62, 63]. However, en-face projections become nonlinearly distorted relative to each other when their curvature relative to the incident beam is significantly different from each other, as in our situation. To circumvent this obstacle, we used 3D volumetric surface point cloud data to volumetrically montage all datasets, which does not suffer the same angular distortion.

Our method for point-cloud-based volumetric montage was inspired by techniques used to address the simultaneous localization and mapping (SLAM) problem, which involves a mobile robot mapping its environment while localizing its position. A conceptual leap in the SLAM problem was that greater correlations between estimates of landmark positions resulted in better solutions to the SLAM problem [64]. In our work, we maximized correlations between landmark points from different OCT volumes to determine the relative transformations between volumes. Additionally, we refined those transformations using the ICP, leveraging the 3-dimensional nature of OCT data. Finally, we performed loop closure using successive iterations of ICP. Concepts such as point cloud mapping, ICP, and loop closure are core to many algorithms to address the SLAM problem. However, our work differs from SLAM, as we do not localize the robot's position relative to the environment or quantify the uncertainty in the robot's pose. Future work will involve localization of the robot's position within the environment and quantifying the uncertainty in the measured transformations and the robot's pose.

In addition to imaging the AHO pathway, robotic OCT can be used for expanded FOV imaging of other tissues, such as the retina [65] or skin. The steps for volumetric montage would be similar, although the choice of landmark points might differ, and the loop closure might not be necessary if the robot's trajectory does not return to the same position during imaging. Moreover, choices in sample arm design may be better suited for imaging larger structures with fewer volumes [66].

As compared with previous demonstrations of robotic OCT, our work differs in the integration of robotic pose information and landmark points to create CIVs with minimal compromise in resolution. Recent demonstrations of increased FOV with robotic OCT were either done *ex vivo*, on homogenous structures where fine alignment of sub-volumes was not critical [18, 23, 24], or did not create full 3D representations [21, 24, 25]. Notably, the limited accuracy of robotic arms was cited as a recurring challenge in creating CIVs [21, 22]. We showed the effectiveness of our reconstruction with 360 renderings and illustrations of the outflow pathway, requiring an accuracy far exceeding the accuracy of the robotic arm

and not capable of being done by registering 2D cross sections. The previous usage of point clouds was limited to generating robotic scan patterns or examining surface morphology [22, 24]. In our robotic OCT, we used point clouds in conjunction with landmark points to merge complex anatomical sub-volumes with heterogeneous structures. In the future, we plan on developing real-time OCT volumetric montaging to reconstruct complex tissue structures.

A crucial step towards real-time volumetric montaging involves automating landmark feature detection and speeding up processing steps. For real-time identification of shared landmark points, features such as SIFT, SURF, ORB, and Harris features have been used to identify matching features between images from other modalities [67, 68]. The analysis of optimal features for OCT imaging is a work in progress. On the hardware side, advances in computation and GPU processing will accelerate the process. Currently, our quasi-Newton algorithm loop closure step is the most time-consuming and thus optional. We found that sequential application of ICP was much faster for loop closure. Another faster alternative includes pose graphs, often used in loop closure in implementing SLAM for other modalities [69]. However, pose graphs do not utilize the full 3D point cloud, decreasing montaging accuracy. A potential compromise is to use pose graphs for real-time loop closure and alternative methods during post-processing. Additionally, the point clouds could be downsampled, speeding up all subsequent steps.

Our goal for this robotic OCT is to translate to human imaging. Most anterior segment OCTs use co-axial imaging because structures like the cornea, iris, and phakic lens are aligned with the optical axis. Because AHO pathways are embedded in the sclera, which attenuates light propagation, conformal illumination becomes critical to minimize optical attenuation. Currently, one clinical solution is to place the OCT on a swing arm, manually swing the device, and ask the person being imaged to take an eccentric view in the opposite direction to achieve normal illumination. However, this operation is not reproducible because patients may not consistently look at the same angle during imaging sessions, and off-axis gazes need to be held by eye muscles, leading to fatigue-related micromovements and motion artifacts. Therefore, we anticipate that robotic OCT will improve AHO pathway assessment as patients can take relaxed, forward-looking views. For human imaging, we need to overcome obstacles caused by, for example, eyelids covering the AHO pathways, implement stricter safety measures, and develop different robotic calibration and scanning protocols.

Finally, we demonstrate the utility of an increased CIV by imaging the full 360 degrees of the AHO pathways, which may improve clinical glaucoma management. From a scientific perspective, reconstructing AHO pathways can improve our understanding of how outflow resistance is regulated. One prevalent question is the root cause of resistance distal to SC [70], which may be caused by pinch points in the distal pathway. We observe collapsed regions of the distal pathway in our resampled CC, which is consistent with the existence of pinch points. From a surgical perspective, it has been demonstrated that segmental patterns in anatomy and outflow may influence the success of MIGS [30, 54, 71–73]. To evaluate this, we quantified segmental differences in SC size and CC distribution, where SC was the largest in the temporal quadrant, and the temporal and nasal quadrants had the most CCs. Although more investigation is necessary to reconcile gold-standard tracer-based imaging of

segmental AHO to segmental anatomy imaged by OCT for guiding surgical decisions, this work lays the technological groundwork for this goal.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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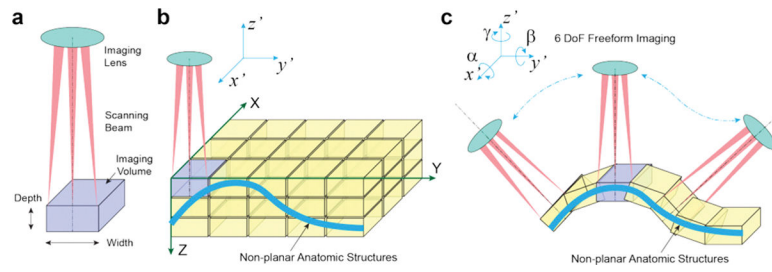
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**Fig. 1.**

Robotic imaging for expanding the optical FOV. a) The optical FOV for conventional microscopy is determined by the DOF. b) Mechanical translation along the XYZ axes can be used to expand the optical FOV. The blue box shows the optical FOV for a single volumetric acquisition, and the yellow boxes show the space in which the FOV can be expanded. c) Robotic scanning increases the FOV and changes the orientation of the sample arm relative to the sample.

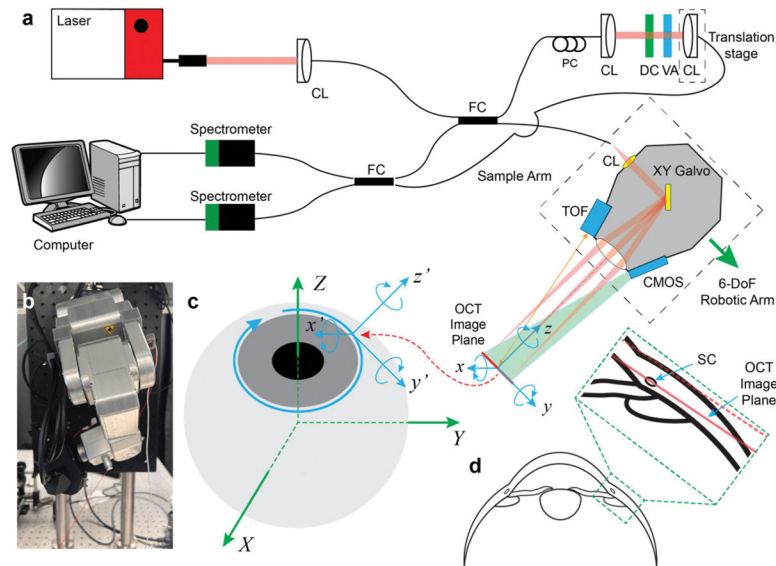


Fig. 2. Experimental setup of robotic vis-OCT and imaging geometry.

a) Schematic of vis-OCT system. CL: collimator; FC: fiber coupler; PC: polarization controller; DC: dispersion compensation; VA: variable attenuator; TOF: time-of-flight sensor; CMOS: pupil camera. The sample arm, inside the dotted black box, is attached to a 6-DoF robotic arm. The imaging coordinates of the OCT are the TRF, with the z -axis parallel to the incident OCT beam and the OCT image plane coinciding with the xy plane of the TRF. b) Photo of 6-DoF robotic arm with sample arm. c) Imaging geometry around the limbus of the eye and the relationship between two reference frames: the ERF represented by the XYZ -coordinate (green) and the TRF of the robot represented by the xyz -coordinate (blue). d) Cross-sectional view of the anterior segment, with a magnified view of the SC shown in the dashed green box.

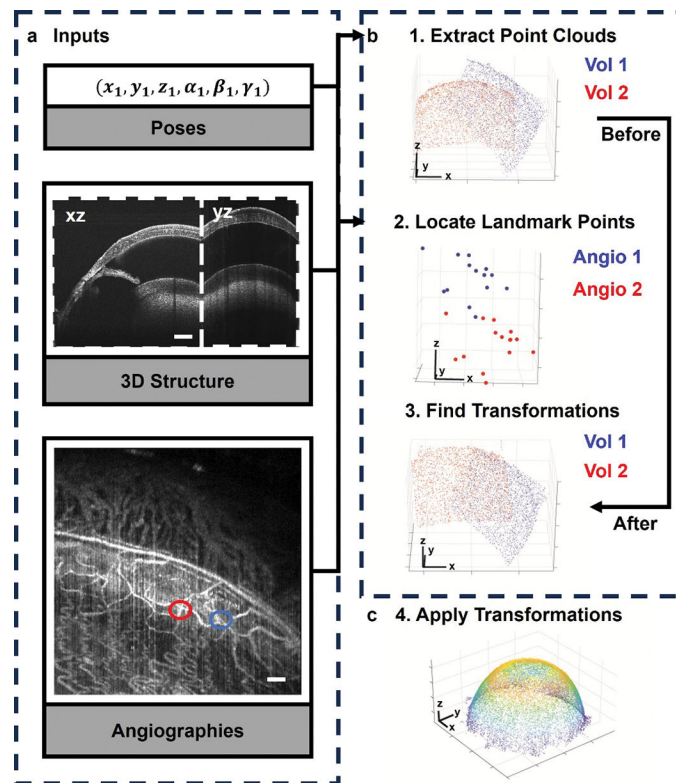


Fig. 3. Workflow of montaging multiple vis-OCT volumes.

a) Input data acquired by robotic OCT. Poses estimate the acquired volume's location and determine which volumes overlap. b) Landmark points from overlapping volumes are used to merge the volumes. (1) The outer surface of the volumetric data is represented as a point cloud; (2) landmark points corresponding to vessel branch points are extracted from the point clouds. Two examples of landmark points are highlighted by the blue and red circles in (a); (3) Common landmark points between volumes are registered to determine the transformation between overlapping volumes. c) All volumes mapped into the ERF. Scale bars for volume data are 200 μm , and for angiographic data are 100 μm .

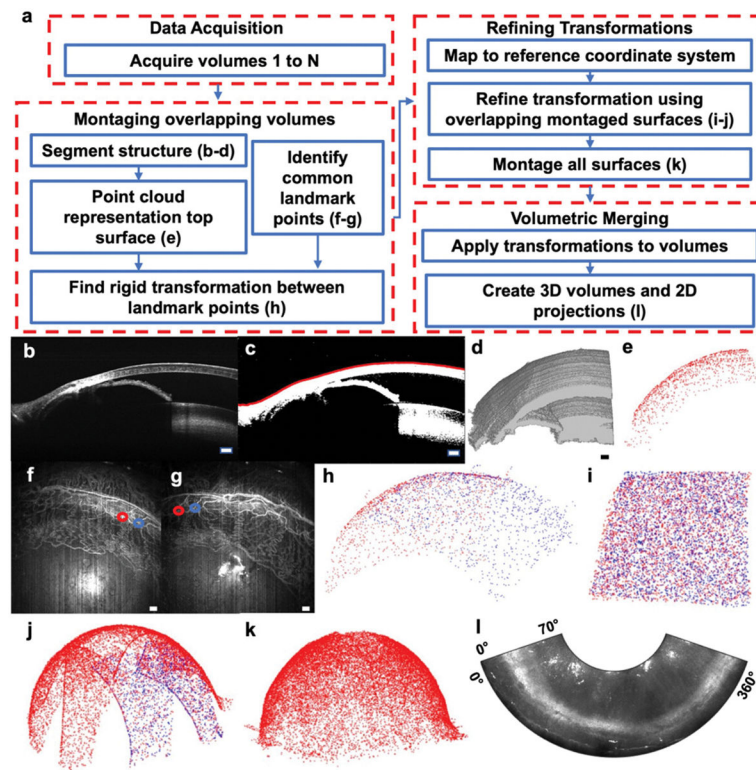


Fig 4. Volumetric registration of OCT volumes.

a) Summary of steps. b) Acquired B-scan. c) B-scan thresholded and binarized, with the outer eye surface marked in red. d) 3D reconstruction of the binarized structure. e) The outer eye surface is represented as a point cloud. f + g) OCTA of overlapping regions. Two pairs of common vessel branch points between the OCTA were marked by blue and red circles. h) Montaging of adjacent surface point clouds. The point cloud of the first volume is shown in red and the point cloud of the second volume is shown in blue. i) Montaging adjacent surface point clouds in the overlapping regions. j) The overlapping region of one point cloud is shown in blue, with all overlapping regions of all other point clouds shown in red. k) Point cloud representation of the entire eye surface. l) Equiconic projection image, with positions along the eye represented as latitude and longitude lines. All point clouds were downsampled by 100 times for visualization purposes. Scale bars are 100 μm .

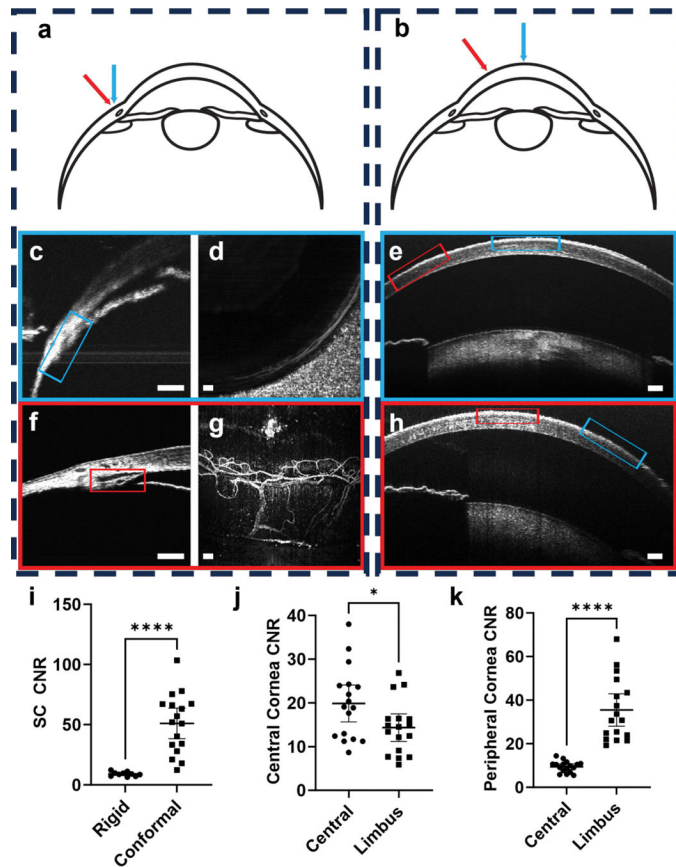


Fig. 5. Effects of vis-OCT illumination angle.

a) Imaging the limbus region at two distinct illumination angles: normal to the central cornea (blue arrow) and normal to the limbus (red arrow). b) Imaging the central cornea at two distinct illumination angles: normal to the central cornea (blue arrow) and normal to the limbus (red arrow). c) B-scan image of SC (blue box) using the illumination angle highlighted by the blue arrow in panel a. d) Vis-OCTA of the limbal vasculature using the illumination angle highlighted by the blue arrow in panel a. e) B-scan image of the cornea using the illumination angle highlighted by the blue arrow in panel b. f) B-scan image of SC (red box) using the illumination angle highlighted by the red arrow in panel a. g) Vis-OCTA of the limbal vasculature using the illumination angle highlighted by the red arrow in panel a. h) B-scan image of the cornea using the illumination angle highlighted by the red arrow in panel b. i) The CNR between SC and surrounding tissue is greater with beam orientated normal to the limbus (conformal) than the central cornea (rigid) ($n = 1$ mouse, 17 B-scans for conformal and 10 B-scans for rigid). j) CNR between the epithelium and stroma in the central cornea is greater when the illumination beam is normal to the central cornea ($n = 1$ mouse, 25 B-scans). k) CNR between the epithelium and stroma in the peripheral cornea is greater when the illumination beam is normal to the limbus ($n = 1$ mouse, 25 B-scans). All scale bars are 100 μm .

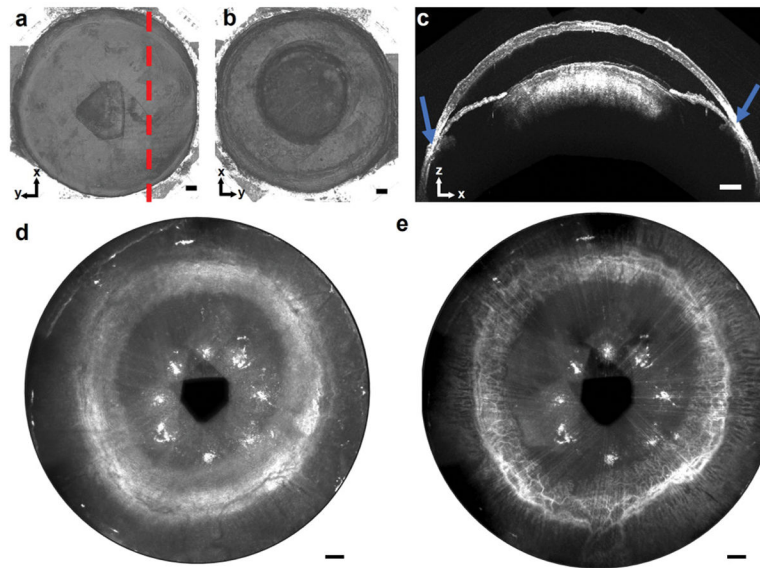


Fig. 6. Reconstructed CIV of the entire anterior segment.

a) Volumetric reconstruction of the anterior segment viewed from the anterior direction, visualizing the outer surfaces of the cornea and limbus. b) Volumetric reconstruction of the anterior segment viewed from the posterior direction, depicting the lens and inner surface of the limbus. c) A representative cross-sectional image along the red line in panel a, with the blue arrows indicating the SC. d) Stereographic projection of structural vis-OCT volumetric data. e) Stereographic projection of the vis-OCTA volumetric data highlighting connected and continuous limbal vasculature (n = 1 mouse). Scale bars are 200 μm .

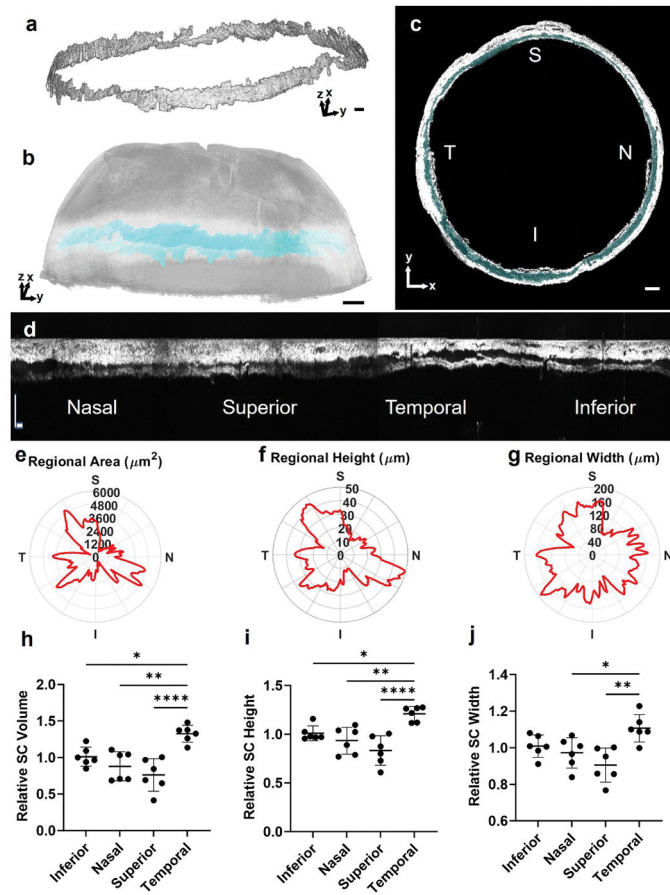


Fig. 7. Circumferentially segmental patterns of SC morphology.

a) A 3D reconstruction of SC. b) SC pseudocolored in light blue within the structural 3D reconstruction of the anterior segment. c) Projection of SC along its major axis circumferentially. The position of SC is overlaid in cyan. d) An unwrapped view of SC along its circumference divided by the four quadrants. e) Polar distribution of SC cross-sectional area as a function of circumferential orientation. f) Polar distribution of SC height as a function of circumferential orientation. g) Polar distribution of SC width as a function of circumferential orientation. h) Relative SC volume for each quadrant compared to the average across all quadrants. i) Relative SC height for each quadrant compared to the average across all quadrants. j) Relative SC width for each quadrant compared to the average across all quadrants (n = 6 mice). * indicates $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. Scale bars in (a)-(c) are 200 μm and (d) is 100 μm .

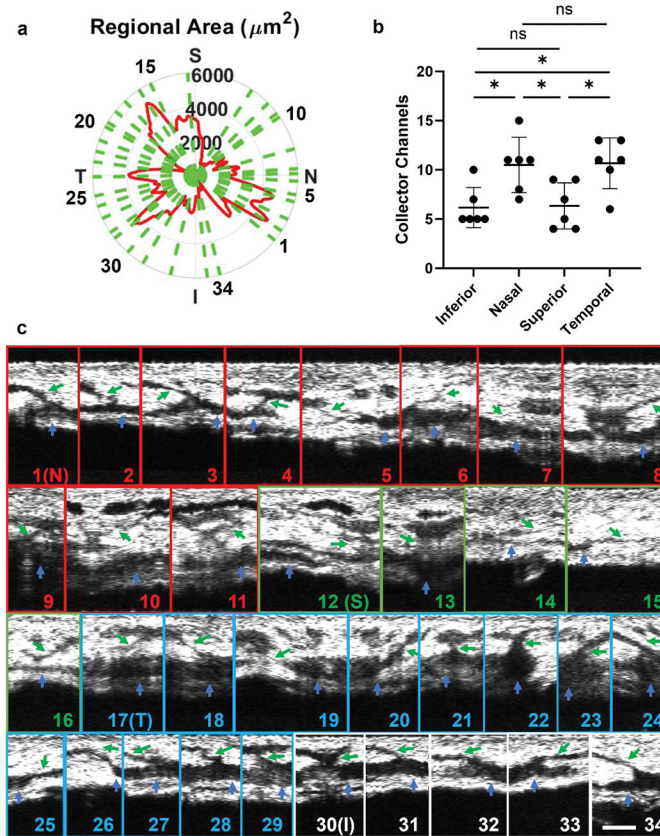


Fig. 8. Circumferentially segmental patterns of CCs.

a) Polar distribution of SC area and CC positions as a function of circumferential orientation in a typical C57BL/6 mouse. The red curve represents the SC area, and the green dotted lines indicate the CC position. b) Distribution of CC per quadrant ($n = 6$ mice). c) Magnified view of all CCs in panel a. The blue arrow points to SC, and the green arrows to CC. The numbering of CC corresponds to the numbering of green lines in panel a. CCs labeled in red are in the nasal quadrant, green in the superior quadrant, blue in the temporal quadrant, and white in the inferior quadrant. The scale bar is 50 μm .

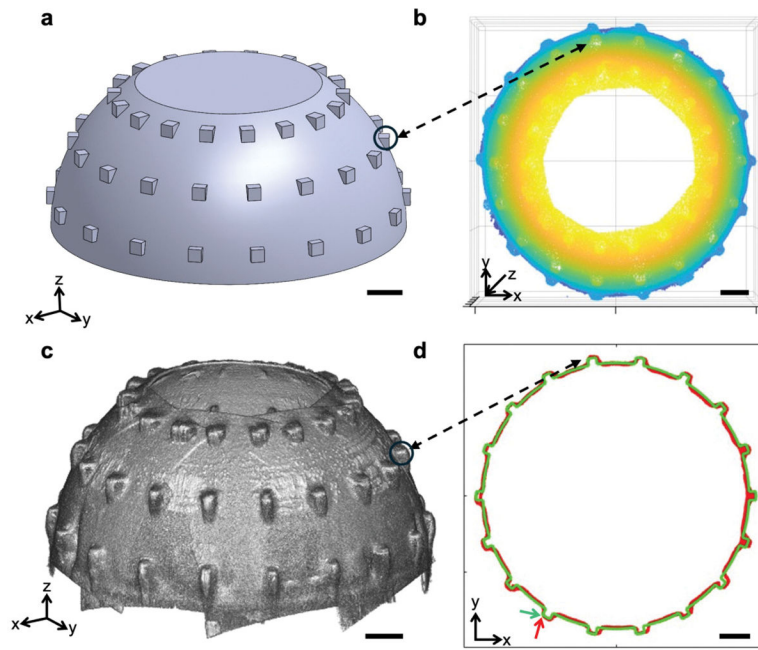


Fig. 9. Imaging a 3D-printed phantom.

(a) Computer-aided design of phantom, with a cuboidal feature within the black circle. (b) Point cloud reconstruction of the phantom, with the black dotted line pointing to the feature circled in panel a. (c) Robotic vis-OCT reconstruction of the phantom, with a cuboidal feature within the black circle. (d) Horizontal cross-section of point cloud at the depth corresponding to the depth of the circled cuboidal feature in panel c. The point cloud is shown in red, while the theoretical phantom design is shown in green. The green arrow points to a vertex of a cuboidal feature, and the red arrow points to the closest point in the point cloud to that feature. Scale bars are 400 μm .

Table 1.

Time consumption of each processing step.

Processing step	Time Usage (Seconds)
Loading data (per volume)	11.2 ± 0.2
Binary segmentation (per volume)	11.4 ± 7.3
Point cloud generation (per volume)	0.8 ± 0.04
Point cloud alignment (per volume pair)	5.2 ± 1.4
Loop closure (sequential ICP)	17.6 ± 3.8
Loop closure (optional, cost function minimization)	641.7 ± 324.4
Mapping data to ERF (per volume)	1.1 ± 0.1